

AEROSP 740 - Aerodynamic Design

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**Final Project: Aerodynamic Design of the successor aircraft
of Shahed-238 Kamikaze Drone**

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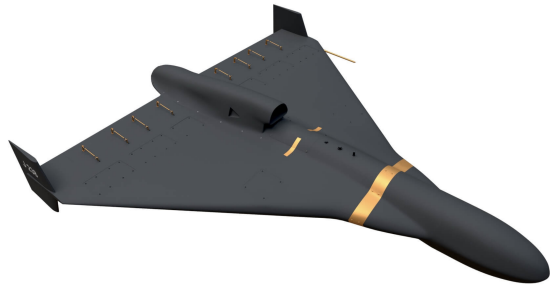
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May 3, 2024

Overview of the Shahed-238

The Shahed-238 (Figures 1a and 1b) is an upgraded Iranian kamikaze drone that was officially introduced in November 2022. It's a modification of the Shahed-136 and is known for its jet propulsion, which likely enhances its speed and operational capabilities. It also incorporates advanced guidance systems, which may include infrared/optical and radar components. The objective of this project is to undertake the aerodynamic design of a successor aircraft of the Shahed-238 in terms of its aerodynamic capabilities to achieve higher speeds at different flight phases and to reduce the overall drag of the aircraft in order to maximize its range and to fly farther ranges in order to maximize its potential as a Kamikaze drone.



(a) A Rendered image of the Shahed-238



(b) Shahed-238

(a) Specifications

The known specifications of the Shahed-238 are as follows:

Specifications	Value
Range	200 km
Cruise Altitude	9000 m
Length	3.5 m
Span	2.5 m
Cruise Speed	550 km/hr ($M = 0.5$)
MTOW	250 kg

Table 1: Known specifications of the Shahed-238

Successor of the Shahed-238

Since the scope of this project is limited to the aerodynamic design of the aircraft, the successor aircraft is assumed to have some target specifications (which are considered to be reasonable improvements for a successor aircraft along the respective tangents) and the aircraft is designed to be able to fulfill these required specifications using **Aerodynamic Design Optimization**.

(a) Designed Specifications

The target specifications of the successor aircraft are given in table 2

Specifications	Value
Range	400 km
Cruise Altitude	9000 m
Length	3.5 m
Span	2.5 m
Cruise Speed	650 km/hr (M = 0.6)
MTOW	350 kg

Table 2: Target specifications for the Successor Aircraft

The takeoff speed of the upgraded aircraft (M = 0.3) is assumed to be at half the mach number as that of the mach number at cruise (M = 0.6) and the Attack / Terminal speed is assumed to be at a slightly lower mach (M = 0.55) than the cruise due to the difference in air density at these altitudes. For this project, we first design the optimal airfoil for the aircraft and then design the optimal planform for a varied set of design freedom given to the optimizer starting from a design closest to the planform of Shahed-238.

Flight Points

Assuming the new MTOW to be 350kg, and the increased range to be 400 kms, the Breguet Range equation 1 is used to find the final weight of the aircraft to determine the required lifts at the cruise and terminal phase:

$$R = \frac{\eta}{SFC} \left(\frac{L}{D} \right) \ln \left(\frac{W_i}{W_f} \right) \quad (1)$$

The Shahed-238 uses the PBSTJ150 as its powerhouse, but little is known about its TSFC and efficiency to the public. The value of $\frac{SFC}{\eta}$ is hence assumed to be around $1.11 * 10^{-5} \text{kg}/(\text{N*s})$, which falls between the typical ranges of the ideal sized turbofan and turbojet engines. The $\frac{L}{D}$ is assumed to be 10, which is achievable feat for Delta winged aircrafts. The final weight (terminal phase) is hence calculated to be 225 kgs, and the mid-cruise weight to be slightly above 285 kgs.

The C_L values for each flight point are hence calculated by assuming $L = W$ in the C_L equation:

$$C_L = \frac{L}{\frac{1}{2}\rho V^2 S} \quad (2)$$

The altitudes of Takeoff and Terminal phases are assumed to be at 100m, considering the high terrains at which military operations are conducted.

(a) Cruise Phase

For the cruise condition, the known parameters are given in table 3:

Parameters	Value
Altitude	9000 m
Mach	0.6
V	182.28 m/s
ρ	$0.47 \text{ kg}/\text{m}^3$
S	4.375 m^2
Mass	285 kg

Table 3: Cruise phase parameters

The required C_L is hence calculated (using 2) to be 0.08.

(b) Takeoff Phase

For the takeoff phase, the known parameters are given in table 4:

Parameters	Value
Altitude	100 m
Mach	0.3
V	101.97 m/s
ρ	1.21 kg/m^3
S	4.375 m^2
Mass	350 kg

Table 4: Takeoff phase parameters

The required C_L is hence calculated (using 2) to be 0.12.

(c) Terminal / Attack Phase

For the Terminal phase, the known parameters are given in table 5:

Parameters	Value
Altitude	100 m
Mach	0.55
V	186.85 m/s
ρ	1.21 kg/m^3
S	4.375 m^2
Mass	225 kg

Table 5: Terminal phase parameters

The required C_L is hence calculated (using 2) to be 0.02. The combined flight point details are given in the table 6:

Parameters	Cruise	Takeoff	Terminal	Units
Altitude	9000	100	100	m
Mach	0.6	0.3	0.55	-
V	182.28	101.97	186.85	m/s
ρ	0.47	1.21	1.21	kg/m^3
Mass	350	285	225	kg
C_L	0.08	0.12	0.02	-

Table 6: Flight point parameters

Multi-point Airfoil Optimization

The airfoil for the successor aircraft is designed to be able to generate the necessary lift at the 3 flight points with minimal overall drag. For the sake of simplicity and to develop a single airfoil section of the aircraft, the aircraft is assumed to be a rectangular wing that is infinitely long to carry out the multipoint airfoil optimization. This assumption allows us to assume the required C_l constraints of the airfoil section to be equal to the C_L s that the aircraft is supposed to produce at the respective flight points.

(a) Starting Airfoil

The optimization is initiated with a **NACA0012** airfoil since we do not know of what shape the multipoint optimizer might want the final airfoil to be and to keep things as simple as possible.

(b) Design Constraints

Since the NACA0012 has a wide crosssection to minimize the drag out of, and the aircraft under consideration does not use a large amount of fuel based on its range and application (calculated to be 125 kg), a 50% volume constraint is given to the optimizer, keeping the thickness constraint at 10% of the initial chord thickness.

The constraints for the C_l for each flight point are provided to the optimizer. (Note: initially, C_m constraints at each flight point followed by a C_m constraint at cruise were given to the optimizer along with the lift constraints but the optimizer was not able to converge with a feasible airfoil for both cases)

(c) Optimization Problem

The optimization problem is defined to reduce the combined drag at all 3 flight points, while having the volume, thickness and C_l constraints as given below:

$$\text{minimize} : (C_{dcruise} + C_{dtakeoff} + C_{dterminal}) \quad (3)$$

$$\text{subject to: } C_{lcruise} = 0.08 \quad (4)$$

$$C_{ltakeoff} = 0.12 \quad (5)$$

$$C_{lterminal} = 0.02 \quad (6)$$

$$\text{while} : Area \geq 0.5Area_{init} \quad (7)$$

$$thickness_{chord} \geq 0.1thickness_{chord} \quad (8)$$

The MACH-Aero framework is used for this optimization.

(c) Optimization Results

The following figure 2 shows the airfoil shapes of the initial and final airfoils:

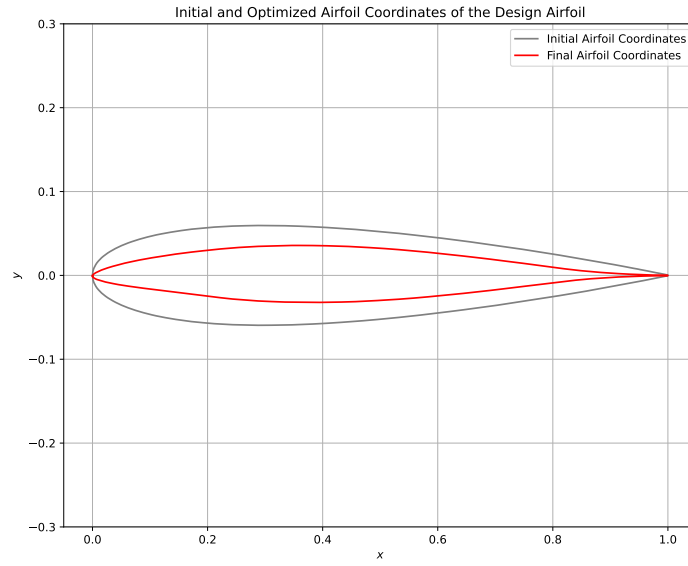
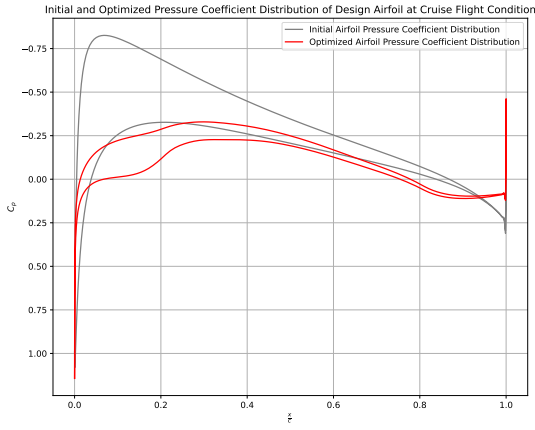
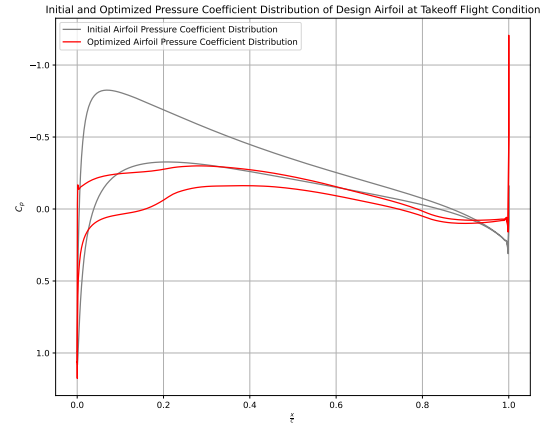


Figure 2: Initial and final airfoil after the optimization

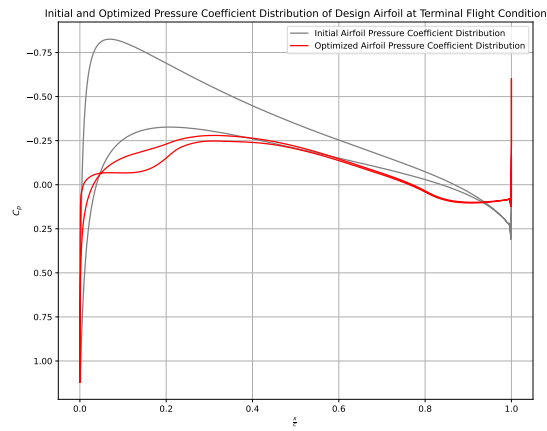
The C_p distribution plots for each flight condition can be seen in figure 3



(a) C_p Distribution for Cruise Flight Condition



(b) C_p Distribution for Takeoff Flight Condition



(c) C_p Distribution for Terminal Flight Condition

Figure 3: C_p distributions for all 3 flight conditions for the optimized airfoil in comparison with the NACA0012

It can be observed that while minimizing the drag, the optimizer reduced the area between the pressure difference between the lower and upper surfaces to constrain itself at the minimal lift that is necessary for our design.

(d) Convergence History

The convergence history can be seen in the figure 4:

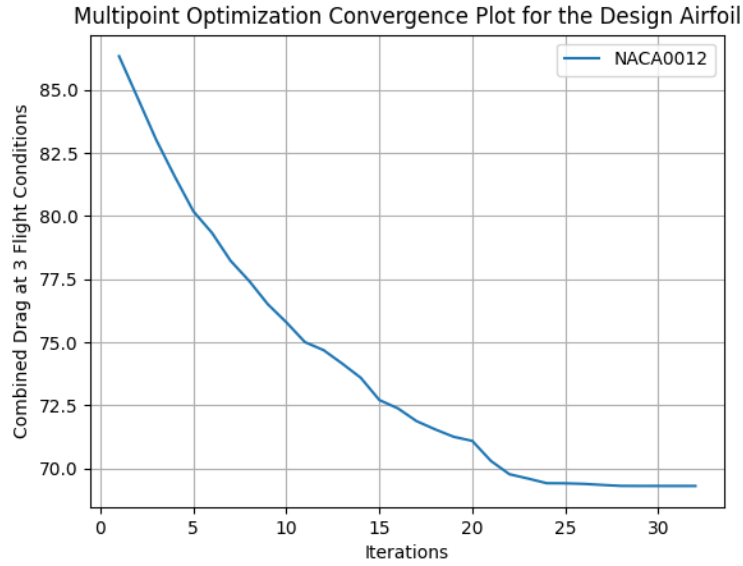


Figure 4: Convergence History of the Multi-point Airfoil Optimization

The optimization resulted in 19.1% of the combined drag for the airfoil section.

Planform Optimization

The aircraft planform is designed using OpenAeroStruct's aerodynamic module, and optimized to fulfill the C_L constraints at the 3 different flight points while allowing it to modify the twist distribution (Cases I, II), taper and the root chord lengths (Case II) in a multi-point optimization. This is followed by giving the optimizer complete design freedom including the sweep and span values to minimize the drag at the cruise condition (Case III).

Starting Planform

The initial planform is given in Figure 10. The specifications are mentioned in table 8

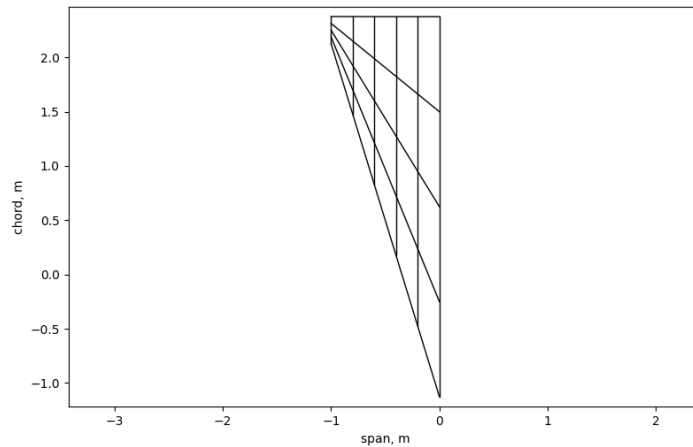


Figure 5: Initial Planform Shape

Parameters	Value
Span	2.5 m
Root Chord	3.5 m
Tip Chord	0.25 m
Twist	-
Sweep	26 degrees

Table 7: Initial planform parameters

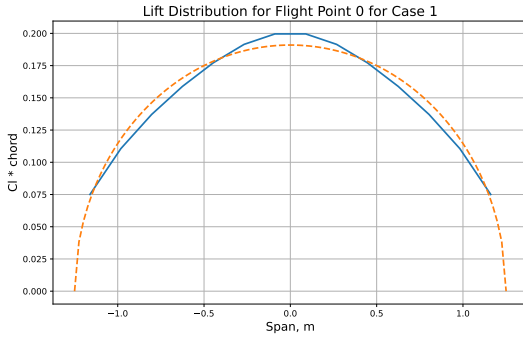
The planform optimization is carried out with 3 cases. The following table describes the cases.

Case	Design Variables	Flight Points
1	Twist	Multipoint
2	Twist, Taper, Root chord	Multipoint
3	Twist, Taper, Root chord, Span, sweep	singlepoint

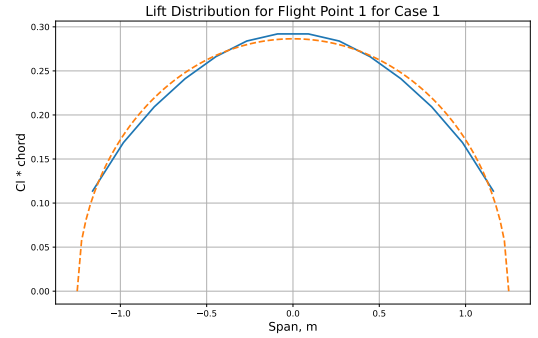
Table 8: Planform optimization cases

Case (I): Twist - Multi-Point

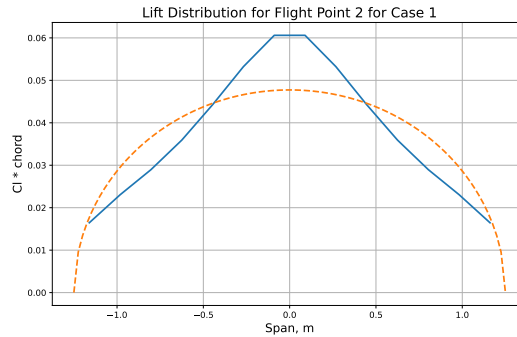
The combined C_D is minimized to 0.069, with the cruise C_D being 0.02188. The spanwise lift distribution for all the flight points is shown in the following figure:



(a) Optimal Spanwise Lift Distribution for Cruise Flight Condition



(b) Optimal Spanwise Lift Distribution for Take-off Flight Condition



(c) Optimal Spanwise Lift Distribution for Terminal Flight Condition

Figure 6: Optimal Spanwise Lift Distribution for all 3 flight conditions for the optimized planform for Case 1

Case (II): Twist & Taper & Root Chord - Multi Point

The combined C_D is minimized to 0.067, with the cruise C_D being 0.02185. The optimal taper is found to be 0.159 and the optimized planform is shown below:

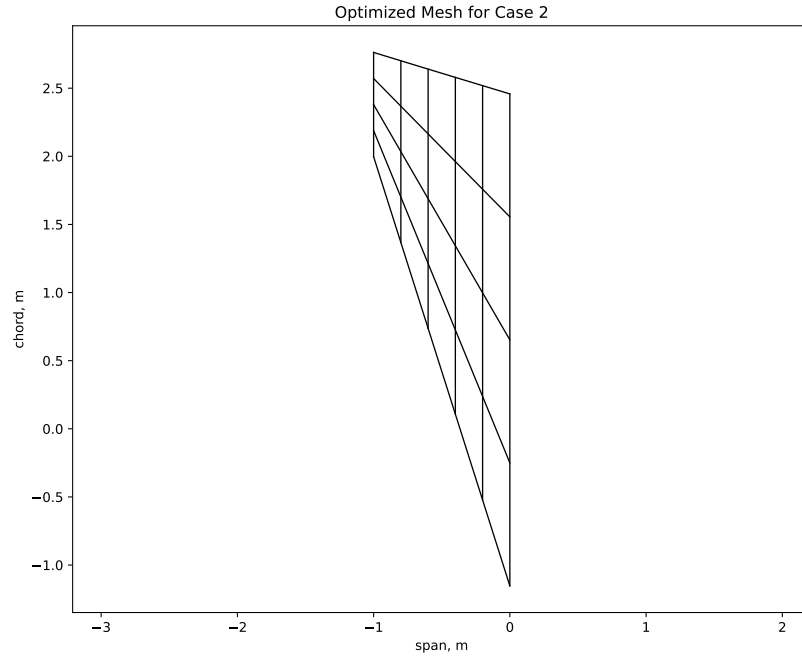
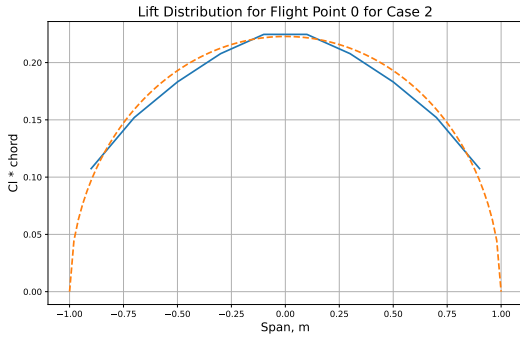
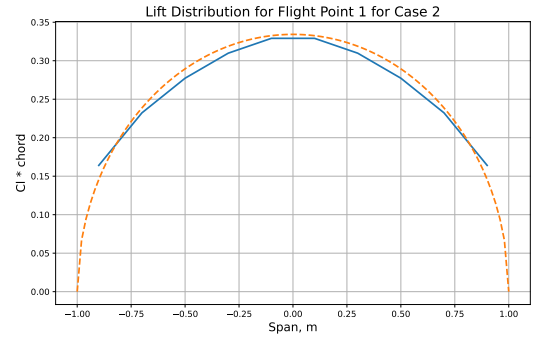


Figure 7: Optimal Planform Shape for Case 2

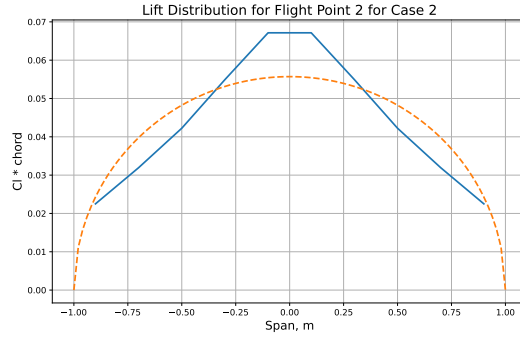
The spanwise lift distribution for all the flight points is shown in the following figure:



(a) Optimal Spanwise Lift Distribution for Cruise Flight Condition



(b) Optimal Spanwise Lift Distribution for Take-off Flight Condition



(c) Optimal Spanwise Lift Distribution for Terminal Flight Condition

Figure 8: Optimal Spanwise Lift Distribution for all 3 flight conditions for the optimized planform for Case 2

Case (III): Complete Design Freedom - Single Point

The cruise C_D is minimized to 0.0228. The optimal taper is found to be 0.30, Optimal sweep of 30° and optimal span of 2m. The optimized planform is shown below:

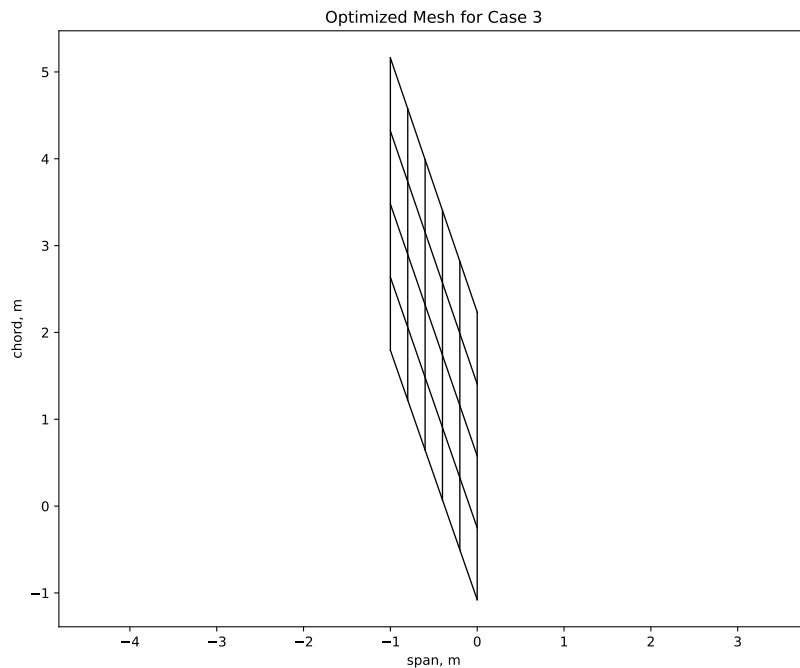


Figure 9: Optimal Planform Shape for Case 3

The optimal lift distribution at cruise is shown below:

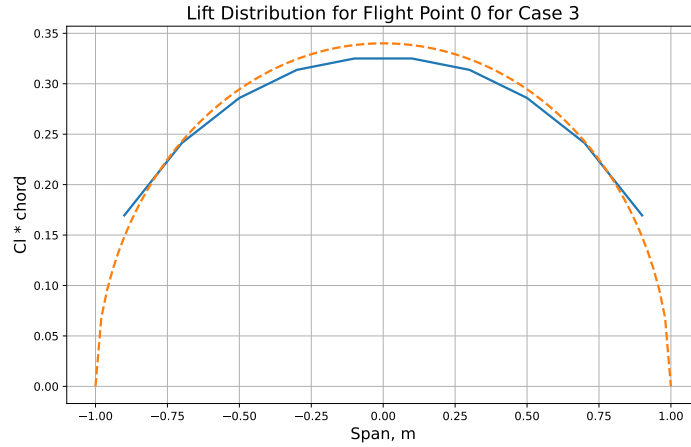


Figure 10: Optimal Spanwise lift distribution at cruise for Case 3

The case 2 proves to be the best overall design, being able to not only minimize the overall drag, but also the drag at cruise.

Conclusion

The successor aircraft of Shahed-238 is hence designed. The Airfoil section is designed using MACH-Aero, followed by the optimal planform shape that is designed using OpenAeroStruct.